SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3 Duration : 1 Hour

Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I B Sandhya(192311292) certify that this submission is my original work
and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:Sandhya

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

Constraints:

i. D1 cannot work Night shift

ii. D2 must work before D3 (Morning < Afternoon < Night)

iii. Only one doctor per shift

iv. D3 cannot work Morning

v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G).

Obstacles block certain paths.

Constraints:

i. Movement allowed: Up, Down, Left, Right

ii. Cost of each move = 1

iii. Cannot pass through obstacles

iv. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.

c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents

 Rationality

 Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

ANSWERS

1. Backtracking Search for Doctor Shift Assignment

Given

Doctors: **D1, D2, D3**

Shifts:

- Morning (M)
- Afternoon (A)
- Night (N)

Constraints

1. D1 cannot work **Night**.
2. D2 must work **before** D3 (Morning < Afternoon < Night).
3. Only one doctor per shift.
4. D3 cannot work **Morning**.
5. Every doctor must be assigned exactly one shift.

Step-by-Step Backtracking

Step 1: Assign D1

Possible shifts for D1:

- Morning ✔
- Afternoon ✔
- Night ✗ (Constraint i)

Choose **D1 → Morning**

Doctor	Shift
D1	Morning

Step 2: Assign D2

Remaining shifts:

- Afternoon
- Night

Try **D2 → Afternoon**

Doctor	Shift
D1	Morning
D2	Afternoon

Constraint Check:

- D2 must be before D3 ✔

Step 3: Assign D3

Remaining shift:

- Night

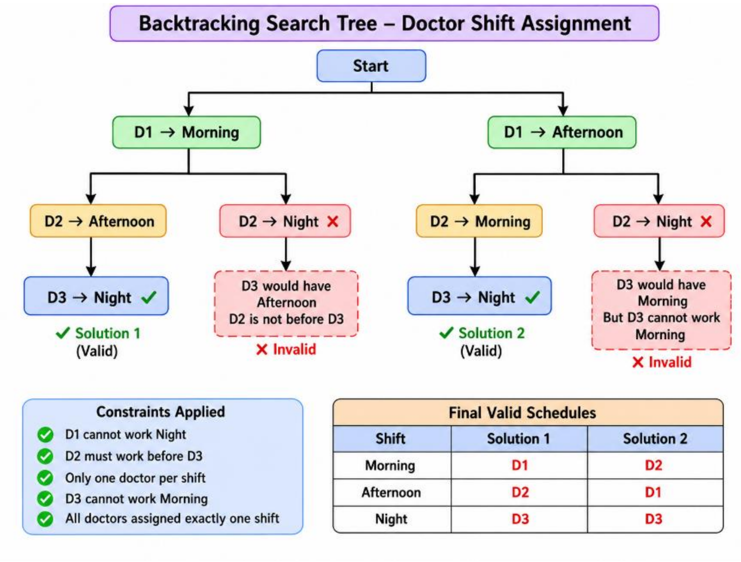
Try **D3 → Night**

Constraint Checks:

- D3 cannot work Morning ✔
- D2 (Afternoon) is before D3 (Night) ✔
- One doctor per shift ✔
- Every doctor assigned exactly one shift ✔

Valid solution found.

Search Tree



Constraint Checks

Constraint	Status
D1 not Night	✔ Satisfied
D2 before D3	✔ Satisfied
One doctor per shift	✔ Satisfied
D3 not Morning	✔ Satisfied
All doctors assigned	✔ Satisfied

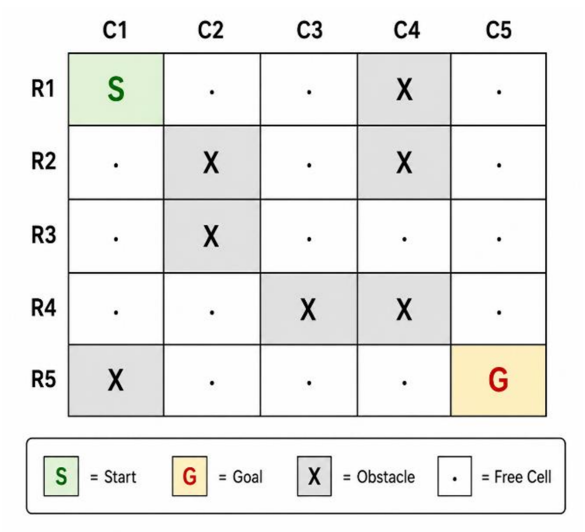
Final Valid Schedule

Shift	Doctor
Morning	D1
Afternoon	D2
Night	D3

Answer:
Morning → D1
Afternoon → D2
Night → D3

2. BFS Search Algorithm (5 × 5 Grid)

Grid Representation



- **Start (S)** = (1,1)
- **Goal (G)** = (5,5)
- **Allowed Moves:** Up, Down, Left, Right
- **Cost per move:** 1
- **Obstacles cannot be crossed**

- **Manhattan Distance:**

$$h(n)=|x-xg|+|y-yg|$$

Step 1: Initial State

Queue:

[(1,1)]

Visited:

{(1,1)}

Step 2: Expand (1,1)

Possible Moves:

- Right → (1,2) ✔
- Down → (2,1) ✔

Queue

[(1,2), (2,1)]

Visited

(1,1), (1,2), (2,1)

Step 3: Expand (1,2)

Neighbors

- Right → (1,3) ✔
- Down → Obstacle ✗
- Left → Already Visited

Queue

[(2,1), (1,3)]

Step 4: Expand (2,1)

Neighbors

- Down → (3,1) ✔
- Right → Obstacle ✗

Queue

[(1,3), (3,1)]

Step 5: Expand (1,3)

Neighbors

- Down → (2,3) ✔
- Right → Obstacle ✗

Queue

[(3,1), (2,3)]

Step 6: Expand (3,1)

Neighbors

- Down → (4,1) ✔

Queue

[(2,3), (4,1)]

Step 7: Expand (2,3)

Neighbors

- Down → (3,3) ✔

Queue

[(4,1), (3,3)]

Step 8: Expand (4,1)

Neighbors

- Right → (4,2) ✔

Queue

[(3,3), (4,2)]

Step 9: Expand (3,3)

Neighbors

- Right → (3,4) ✔

Queue

[(4,2), (3,4)]

Step 10: Expand (4,2)

Neighbors

- Down → (5,2) ✔

Queue

[(3,4), (5,2)]

Step 11: Expand (3,4)

Neighbors

- Right → (3,5) ✔

Queue

[(5,2), (3,5)]

Step 12: Expand (5,2)

Neighbors

- Right → (5,3) ✔

Queue

[(3,5), (5,3)]

Step 13: Expand (3,5)

Neighbors

- Down → (4,5) ✔

Queue

[(5,3), (4,5)]

Step 14: Expand (5,3)

Neighbors

- Right → (5,4) ✔

Queue

[(5,4), (5,5)]

Step 15: Expand (4,5)

Neighbors

- Down → (5,5) ✔ Goal Found

Queue

[(5,4), (5,5)]

Manhattan Distance Values

Node	h(n)
(1,1)	8
(1,2)	7
(1,3)	6
(2,3)	5
(3,3)	4
(3,4)	3
(3,5)	2
(4,5)	1
(5,5)	0

Optimal Path Found by BFS

```
S → (1,2) → (1,3) → (2,3) → (3,3) → (3,4) → (3,5) → (4,5) → G
```

Total Path Cost

- Number of moves = **8**
- Cost per move = **1**

Total Cost = 8 × 1 = 8

Final Answer

- **Search Algorithm:** Breadth-First Search (BFS)

- **Heuristic Used:** Manhattan Distance (shown for reference)
- **Optimal Path:** `S → (1,2) → (1,3) → (2,3) → (3,3) → (3,4) → (3,5) → (4,5) → G`
- **Total Cost: 8**

3. Autonomous Rescue Robot in a Disaster-Affected Building

(a) Analysis of Constraints and Their Effect on Decision Making

Constraint	Effect on Robot's Decision
Robot can move Up, Down, Left, Right	The robot explores only four possible directions from each room.
Obstacles cannot be traversed	The robot must avoid blocked rooms and find an alternate route.
Limited sensing (adjacent cells only)	The robot does not know the entire building initially and discovers new information while moving.
Cost of each move = 1	
Risky zones have additional cost of 2	The robot prefers safer paths even if they are slightly longer, minimizing total cost.
Minimize total cost	The robot chooses the least-cost path rather than the shortest path alone.
Dynamic knowledge update (Online Search)	The robot continuously updates its map as it discovers new rooms or blocked paths.

(b) Solution Approach (Uniform Cost Search + Online Search Agent)

Step 1: Initialize

- Start at **S**.
- Set total cost = **0**.
- Create an empty map of explored rooms.

Step 2: Sense Environment

- Observe adjacent cells.

- Identify free rooms, blocked rooms, and risky zones.

Step 3: Expand Possible Moves

- Generate valid neighboring cells.
- Ignore blocked cells.

Step 4: Calculate Cost

- Normal room = **+1**
- Risky room = **+3** (1 movement + 2 risk)

Step 5: Select Lowest-Cost Path

- Use **Uniform Cost Search (Priority Queue)**.
- Expand the path with the minimum cumulative cost.

Step 6: Update Knowledge

- Add newly discovered rooms to the internal map.
- If a blocked path is found, recalculate the route.

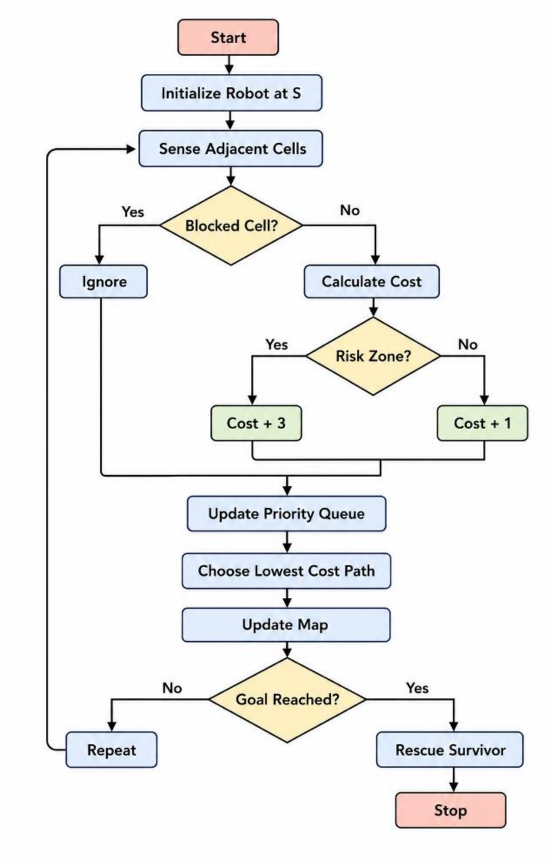
Step 7: Repeat

- Continue until the survivor (**G**) is reached.

Step 8: Rescue Survivor

- Stop after reaching **G** with the minimum total cost.

Flowchart



(c) AI Concepts Applied

1. Intelligent Agent

- The rescue robot acts as an **intelligent agent**.
- It senses the environment using sensors.
- It decides the best action.
- It moves safely toward the survivor.

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2. Rationality

The robot is **rational** because it:

- Chooses the least-cost path.
- Avoids obstacles.
- Avoids risky areas whenever possible.
- Updates decisions based on new information.

3. Environment Type

Property	Type
Observability	Partially Observable

Property	Type
Determinism	Deterministic
Dynamics	Dynamic (environment knowledge changes during exploration)
Agent Type	Single Agent
State Space	Discrete
Decision Making	Sequential

(d) Justification of the Chosen Approach

Uniform Cost Search (UCS) combined with an **Online Search Agent** is the most suitable approach because:

- UCS always finds the **minimum-cost path**.
- It considers both movement cost and additional risk cost.
- Online Search enables the robot to work with **limited sensing**.
- The robot can **re-plan** whenever new obstacles are discovered.
- The approach ensures **safe, optimal, and adaptive navigation** in an unknown disaster environment.